

**Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of
Selective Interference**

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Abstract

In the trauma-film paradigm, visuospatial tasks performed after film encoding or a later film reminder can reduce intrusive memories while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that intrusive and voluntary remembering depend on differently vulnerable memory representations. Here we simulate an alternative retrieved-context account in an emotional Context Maintenance and Retrieval (eCMR) model, in which unguided retrieval and deliberate recall draw on the same film and task associations but differ in how strongly retrieval is constrained toward film items. In the model, film context remains active after encoding and can later be reinstated by reminders, so an interfering task performed in either state associates task items with context states similar to film context. At retrieval, overlapping context can cue both film items and task competitors, reducing the probability that film items win retrieval even though stored film-item associations remain unchanged. Deliberate recall is less vulnerable when control settings reinstate film-associated context, sharpen selection among candidates cued by context, and limit how much off-target samples redirect later search. Test-provided film cues can likewise update context and reduce reliance on context-guided selection. Together, the simulations reproduce selective impairment of unguided retrieval relative to deliberate recall and specify when the intrusion-voluntary-memory dissociation should be more or less pronounced, helping to interpret heterogeneity across empirical paradigms. Our results show that selective interference can arise from retrieval dynamics in a single retrieved-context system.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of Selective Interference

Introduction

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. A rational analysis of episodic memory suggests that highly consequential events should be retained well enough for action-relevant details to inform later evaluation, planning, and choice (Boyle & Brown, 2025; Zhou et al., 2025). At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current concerns. Intrusive memories after trauma suggest that durable encoding can outpace regulatory control. In PTSD, the trauma event returns without being deliberately sought and without being well matched to the demands of the present (Brewin et al., 2010; Ehlers & Clark, 2000; Iyadurai et al., 2019; Krans et al., 2009). Understanding why such memories arise, and how they differ from voluntary memory for the same event, is therefore a shared target for both clinical science and mechanistic theories of memory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event (Holmes & Bourne, 2008; James et al., 2016). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to return involuntarily. In such research, voluntary memory can be relatively preserved, and its relationship to intrusion frequency can be weak, such that the two measures dissociate (Holmes et al., 2004; James et al., 2016). The same encoded episode can therefore support both involuntary and deliberate access, and these modes can come apart empirically.

Visuospatial interference tasks make this dissociation experimentally sharper. When brief tasks such as Tetris are performed after film encoding, subsequent intrusions can be reduced while voluntary memory for the same material shows little measurable impairment (Deepröse et al., 2012; Holmes et al., 2009, 2010; Lau-Zhu et al., 2019). In delayed variants, participants return after about 24 hours to 3 days, receive visual reminder cues for the film, and then complete Tetris;

these reminder-plus-Tetris procedures can similarly reduce subsequent intrusions ([James et al., 2015](#); [Kessler et al., 2020](#)). Similar procedures have also been extended to real-life traumatic events, including motor vehicle accidents, emergency caesarean section, and war-related trauma ([Horsch et al., 2017](#); [Iyadurai et al., 2018](#); [Kanstrup et al., 2021](#)). Two recent meta-analyses support effects of cognitive interventions on intrusive memories, although the evidence base is heterogeneous and recent null findings caution against treating the effect as settled or uniform ([Asselbergs et al., 2023](#); [Varma et al., 2024](#); [Wessel et al., 2025](#)). This selective interference effect therefore raises a theoretical question: when and why can a post-encoding or post-reminder manipulation alter one mode of retrieval more than another, even while both draw on the same recent experience?

One prominent family of interpretations treats the selective interference effect as evidence that intrusion-supporting memory representations are more vulnerable to interference than representations supporting deliberate recall ([Brewin et al., 1996](#); [Brewin, 2014](#)). Dual-representation accounts make this claim architecturally: traumatic experience is proposed to produce distinct sensory and contextual memory traces, with the first supporting involuntary re-experiencing and the second supporting deliberate recall ([Brewin et al., 2010](#); [Brewin, 2014](#)). In immediate post-film interventions, a visuospatial task is then understood to draw on working-memory resources needed to maintain and consolidate sensory-perceptual imagery, reducing intrusion frequency while voluntary-memory measures remain relatively preserved ([Baddeley & Andrade, 2000](#); [Deepröse et al., 2012](#); [Holmes et al., 2009, 2010](#); [Lau-Zhu et al., 2019](#)). For post-reminder interventions, related interpretations often appeal to reconsolidation or post-reactivation restabilization: the reminder reactivates the relevant trace, and the interfering task disrupts its later availability ([James et al., 2015](#); [Kindt et al., 2009](#)). Across these variants, selective interference is explained by treating the manipulation as acting more strongly on intrusion-supporting or reactivated memory than on the basis for deliberate recall.

Here we argue that selective interference need not be interpreted as evidence for differently vulnerable memory representations. It may instead reflect differences in retrieval

processes operating over a shared set of episodic representations. Mechanisms supporting selective, goal-directed retrieval in voluntary recall may be sufficient to sidestep or overcome interference that nonetheless reduces the likelihood of unintentional recall. On this interpretation, the selective interference effect reflects a capacity to intentionally focus retrieval on goal-relevant event information while reducing the influence of competing memories.

Formal modeling is useful here because the verbal dissociation leaves several commitments underspecified. Computational accounts of emotional memory have begun to formalize how emotional-context retrieval dynamics can produce intrusions and how reminder cues and intervention strength alter the persistence of intrusive memories ([Bonsall & Holmes, 2023](#); [Cohen & Kahana, 2022](#)). The present account centers on the selective interference effect: why the same film and interference manipulation can produce different effects across unguided and deliberate retrieval measures. Explaining that dissociation requires specifying whether the task weakens film memory or adds competitors at retrieval, when those competitors are cued alongside film items, and how retrieval instructions or supplied cues change the outcome. A model can therefore turn the retrieval-process interpretation into predictions about when selective interference should appear, rather than only redescribing the dissociation after the fact.

We formalize this proposal in an emotional variant ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)) of the Context Maintenance and Retrieval (CMR) framework ([Polyn et al., 2009a](#)). Retrieved-context models treat context as a gradually changing state that is shaped by experience, reinstated by retrieval, and used to cue later memory search ([Howard & Kahana, 2002](#); [Polyn et al., 2009a](#); [Sederberg et al., 2008](#)). In CMR, items and context are linked by bidirectional associations: context cues candidate items, and retrieved or supplied items reinstate associated context ([Polyn et al., 2009a, 2009b](#)).

Controlled retrieval can be represented in CMR in several ways. A retrieval goal can set the starting context for search by reinstating context associated with the target episode, rather than beginning from whatever context happens to be active at test. Control can also make the current context cue more decisive, so that candidates strongly supported by that context are more likely to

win retrieval than weaker competitors. Finally, monitoring can evaluate sampled candidates against the retrieval goal, limiting how much off-target items shape subsequent search. Prior retrieved-context models introduced related mechanisms to explain recall from goal-relevant context, differences in how selectively items are sampled, and the ability to keep retrieval focused on a target list (Lohnas et al., 2015; Polyn et al., 2009a; Sederberg et al., 2008). Emotional CMR variants can preserve these retrieved-context mechanisms while adding an emotional-context pathway and emotion-dependent binding (Cohen & Kahana, 2022; Talmi et al., 2019). Together, these properties make CMR a principled setting for a single-system retrieved-context account of selective interference.

To map this framework onto the trauma-film paradigm, the model presents a film sequence, an intervening delay, film reminder cues, an interference task, and a post-task delay while context evolves. In the simulations, the intervening and post-task delays are implemented as non-film filler items, so context continues to drift between film encoding, reminder, task encoding, and retrieval. During the reminder phase, film reminder cues can reinstate film-associated context before task encoding, so task items are encoded while context remains similar to states associated with film items. At retrieval, the current context state cues memory search, so film and task items receive retrieval support to the extent that they have been associated with similar context states. Unguided and deliberate retrieval can draw on the same learned associations. Unguided retrieval begins from the current context, which can support film and task candidates together, whereas deliberate recall can use control settings to orient search toward film targets. A selective interference effect in this model would therefore reflect differential control over a shared associative memory, rather than selective vulnerability of separate memory representations.

The simulations develop the retrieved-context account through five linked demonstrations. Simulation 1 shows how a task performed after a film reminder can become a retrieval competitor and reduce later film recall. Simulation 2 decomposes deliberate recall into control settings that can reduce this competition by changing where retrieval begins, how selectively candidates are sampled, and how much off-target samples shape ongoing search. The next simulations examine a

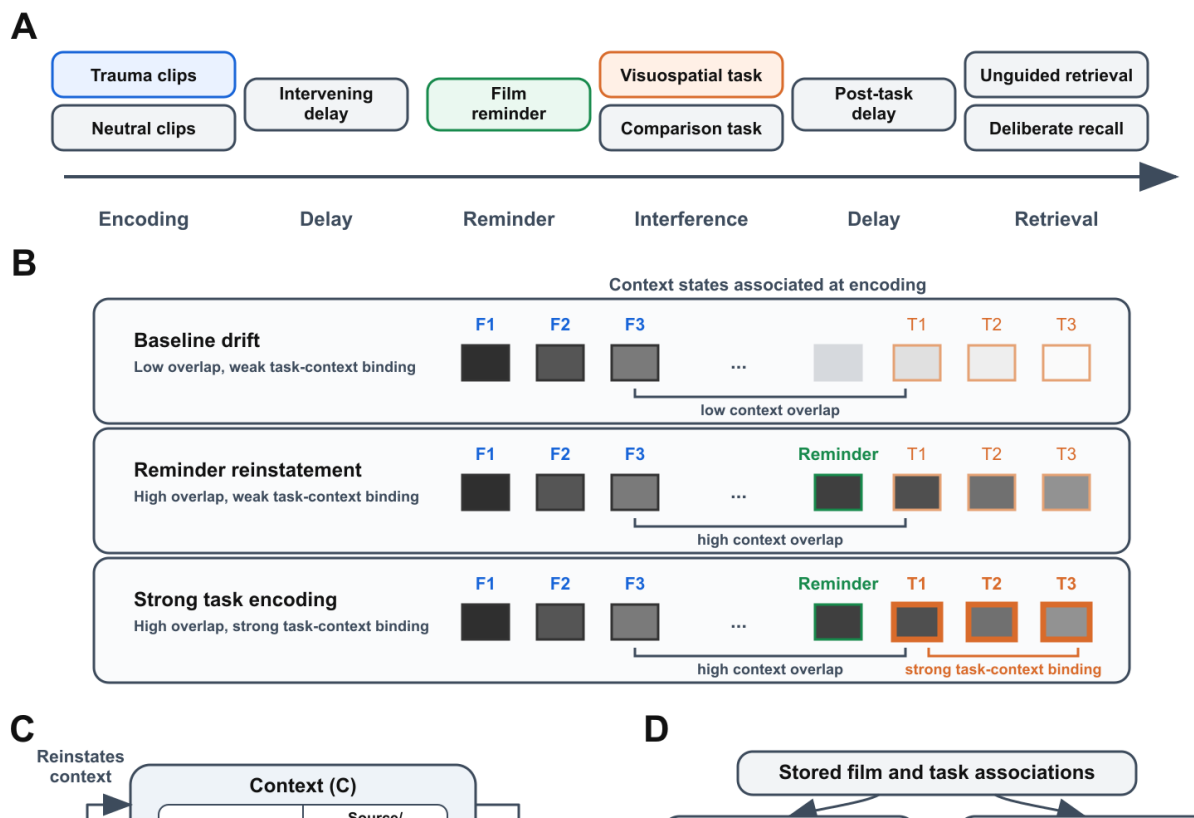
consequence of this retrieval interpretation: the effect should depend on how the memory test supplies film information. Simulation 3 adds test-phase film cues during retrieval, testing whether such cues improve film recall without necessarily removing task competition. Simulation 4 considers recognition-like tests in which the film item is supplied, reducing the need for that item to win a search guided by context. Simulation 5 explores which task properties make interference stronger or weaker. Together, the simulations specify a retrieval mechanism for the selective interference effect and derive predictions about when the dissociation between intrusive and deliberate memory should appear across paradigms.

A Retrieved-Context Account of Selective Interference

The simulations rely on three retrieved-context mechanisms. First, current context cues candidate items, so film and task items can compete when they have been encoded in similar context states. Second, retrieved, reminded, or supplied items reinstate associated context, allowing reminders and test cues to change the context that guides later retrieval. These two item-context mechanisms are standard components of retrieved-context models ([Polyn et al., 2009a](#); [Sederberg et al., 2008](#)). Third, controlled retrieval can alter this same retrieval process. This control component changes three settings on the shared retrieval process: the context state from which search begins, how strongly the best-supported candidates are favored during selection, and how much off-target samples update later context. These settings correspond to start-of-list context reinstatement, retrieval selectivity, and target monitoring in later CMR applications ([Lohnas et al., 2015](#)). Figure 1 shows how these mechanisms map onto the trauma-film paradigm.

Figure 1

A retrieved-context account of selective interference. (A) The model represents the paradigm as film encoding, an intervening delay, film reminder cues, an interference or comparison task, a post-task delay, and retrieval tests. (B) Film items (F1-F3) and task items (T1-T3) are associated with the context states active when they are encoded, shown as gray squares. Reminder cues increase overlap between the film context trajectory and the context states active during task encoding. Strong task encoding strengthens the binding between task items and those overlapping states. Together, overlap and binding make task items more likely to compete with film items at retrieval. (C) Context-to-item associations cue candidate items, and item-to-context associations let recalled, reminded, or supplied items reinstate associated context. (D) In unguided retrieval, ongoing context can cue film items and task competitors together. In deliberate recall, start-of-list context reinstatement increases film support, and retrieval selectivity makes that support advantage more consequential while task competitors remain available. Blue marks film items or film support, green marks reminder cues, and gray marks neutral, delay, comparison, or context-state information. Orange marks task items, task-associated context states, and task support. Dashed orange arrows and outlines mark task competitors that are still available but less favored.



Contextual similarity is established during encoding. We use item in the CMR sense to mean a discrete encoded unit, not the whole film: a film item can stand for a scene or segment, and task or filler items stand for events during the task or delay. Film items and task items are distinct item representations, but each is linked to the context state active when it is encountered. Task items are most likely to share film-associated temporal context when they are encoded soon after the film or after a reminder has reinstated film-associated context. Task items can also share film-associated source context when they share a non-temporal source feature with film items. Emotional extensions of CMR treat affect as one source-context dimension ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)), so shared affective or source features can be represented as shared source context. Delay periods, filled either by laboratory tasks or by later daily activity, move context away from the film, reducing the likelihood that film items and task competitors enter the same later retrieval competition. These conditions determine whether film and task items enter the same retrieval competition; task encoding strength determines how strongly task items compete in that competition. In the present model, shared source context can amplify competition when film and task items share the same source feature.

Context-to-item retrieval uses the current context state to cue candidate items. Unguided retrieval approximates the retrieval condition relevant to intrusive or involuntary access: film memories can be sampled without an explicit goal to recall the film and without film cues supplied by the test. When the current context resembles states associated with both film items and task items, it supports candidates from both phases. This mixed cue makes film recall vulnerable to task competition: as task items receive support, any given film item becomes less likely to be selected over a competitor. The amount of competition depends on how many task competitors are available, how strongly those competitors were encoded, and how similar their temporal or source context is to film context. In these simulations, task modality is represented through task encoding strength and the similarity between task-associated context and film-associated context.

Item-to-context associations allow recalled items, reminder cues, and supplied film items to reinstate associated context. In ordinary recall, a retrieved item shifts context toward the state in

which that item was encoded, making nearby studied items more likely to be retrieved next. This context-updating loop is the standard retrieved-context explanation of temporal contiguity and temporally organized recall ([Howard & Kahana, 2002](#); [Kahana, 1996](#); [Sederberg et al., 2008](#)). The same dynamics can persist across filled delays, allowing recent but irrelevant experiences to intrude during recall of a target episode, with intrusion frequency decaying gradually over intervening time ([Howard et al., 2008](#); [Lohnas et al., 2015](#); [Zaromb et al., 2006](#)). When the interference task is delayed after film encoding, studies have presented a film reminder before the task and interpreted the procedure as memory reactivation ([James et al., 2015](#); [Kessler et al., 2020](#)). On the present account, the reminder uses item-to-context associations to reinstate film-associated context, paralleling temporal-context explanations of human post-reminder effects ([Sederberg et al., 2011](#)). The effect of a delayed interference task is therefore graded by reminder strength and subsequent drift: stronger reminder reinstatement and weaker subsequent drift leave more film-associated context available when the task is encoded.

Item-to-context associations also matter at test. During recall or monitoring, retrieved or test-presented items update context as retrieval unfolds. Retrieving a film item pulls context toward film-associated states and can support retrieval of additional film items. When a task competitor is sampled, it pulls context toward task-associated states and can prolong competition. Film-related cues provided by the test can likewise update context, making subsequent retrieval less dependent on maintaining film context internally. More generally, episodic-updating studies show that returning to the original learning context can itself function as a reminder, reactivating a prior episode and influencing whether later material is linked to it ([Hupbach et al., 2008](#)). Retrieval tests that supply a film item can be less vulnerable to interference, because the supplied item reinstates associated context and reduces reliance on selecting film items from a mixed competitor set ([Lau-Zhu et al., 2021](#)).

Deliberate recall is represented as controlled retrieval over the same stored associations used by unguided retrieval. Protection of film recall depends on how these control settings change context-to-item search. Like unguided retrieval, deliberate recall can still be guided by context

states that cue both film items and task competitors, so it remains exposed to interference. The first control setting is start-of-list context reinstatement, which drifts retrieval context toward the start of the encoded sequence before recall. Because the film is encoded first in the present paradigm, that start state supports early film items more than later task or filler items. The second is retrieval selectivity, which makes differences in current context-to-item support have stronger consequences for selection. This setting favors film items only when the current retrieval context already supports them over competitors. The third is target monitoring, which limits how much sampled off-target items update later context. Together, these settings provide graded protection: strong competition can still reduce deliberate film recall, but less than unguided retrieval from a context that supports both film and task candidates.

The simulations instantiate these commitments in an emotional CMR model (Cohen & Kahana, 2022; Talmi et al., 2019). The model departs from standard list-learning applications of CMR in two ways. First, it treats the film sequence, intervening delay, reminder, post-film task, and post-task delay as one continuous stream through which context evolves. Within that stream, reminder cues can reinstate film-associated context before task encoding. Second, it simulates non-film material from the delay and task periods as distinct items that can be sampled during retrieval and compete with film items. Because those items can win retrieval attempts, the model can express selective interference as a shift in what is retrieved rather than as weakening of film-item associations.

Model Specification

The formal model keeps the standard CMR separation between item features, context, feature-to-context memory, and context-to-feature memory. The section first defines these inherited representations and base parameters, then specifies the context-updating, learning, retrieval, and control operations used to configure the selective interference simulations.

Representations and Parameters

Each study event i has an orthonormal feature vector $f_i \in \mathbb{R}^N$, where N is the number of encoded item features in the simulated paradigm. The model maintains a unit-length context

vector $c_i \in \mathbb{R}^D$, where D is the number of context units. The core context units represent the start state and item-associated temporal context. Before encoding begins, context is initialized to the start state c_0 , with the start unit active and item-associated temporal-context units inactive. Subsequent events update this state, so the context associated with each film, task, or filler item depends on where it occurs in the continuous paradigm stream. The model contains two associative matrices. The matrix $M^{FC} \in \mathbb{R}^{D \times N}$ maps feature vectors to context input. The matrix $M^{CF} \in \mathbb{R}^{N \times D}$ maps context to feature support. All vectors are treated as column vectors, so $M^{FC}f$ is a context vector and $M^{CF}c$ is a feature-support vector.

Table 1 lists the model parameters used in the equations below. The notation is adapted across several retrieved-context and CMR implementations, including foundational free-recall models (Polyn et al., 2009a; Sederberg et al., 2008), later likelihood-based variants (Kragel et al., 2015; Morton & Polyn, 2016), emotional CMR variants (Cohen & Kahana, 2022; Talmi et al., 2019), and target-monitoring extensions (Lohnas et al., 2015).

Pre-experimental associations represent background memory structure already present before the simulated paradigm begins. They give each item baseline self-support and shared support, so later retrieval reflects both pre-existing structure and associations learned during the film, task, and filler phases. The selective interference simulations vary the new associations formed across phases. These baseline associations are fixed model architecture and are held constant across the selective interference simulations. For the start unit and item-associated temporal context units:

$$M_{pre}^{FC}(j, i) = \begin{cases} 1 - \gamma & \text{if } j = i + 1 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$M_{pre}^{CF}(i, j) = \begin{cases} 0 & \text{if } j = 0 \\ \delta & \text{if } j = i + 1 \\ \alpha & \text{if } j > 0 \text{ and } j \neq i + 1 \end{cases} \quad (2)$$

The start context has no pre-experimental context-to-feature support. For context units corresponding to items, δ gives each item self-support and α gives all other items shared support. This initialization supplies baseline support before experimental learning.

Context Updating, Learning, and Retrieval

Context updating defines both gradual drift during encoding and reinstatement by recalled, reminded, or supplied items. The same feature-to-context operation updates context when an item is encoded, recalled, or supplied as a cue. When feature f_i is active, it probes feature-to-context memory to retrieve context input:

$$c_i^{\text{IN}} = M^{FC} f_i \quad (3)$$

The model normalizes this input to unit length, yielding $\hat{c}_i^{\text{IN}}$. Context then drifts toward the normalized input with a drift rate β :

$$c_i = \rho_i c_{i-1} + \beta \hat{c}_i^{\text{IN}} \quad (4)$$

Let $d_i = c_{i-1} \cdot \hat{c}_i^{\text{IN}}$. The normalization coefficient is:

$$\rho_i = \sqrt{1 + \beta^2 (d_i^2 - 1)} - \beta d_i \quad (5)$$

The coefficient ρ_i is the nonnegative solution that keeps c_i at unit length. Associative learning defines how encoded film, task, and filler events become later retrieval candidates. When a new event is encoded, the active feature and updated context are associated via outer-product learning. Feature-to-context associations update as:

$$\Delta M^{FC} = \gamma c_i f_i^{\text{T}} \quad (6)$$

A primacy function weights context-to-feature learning:

$$\phi_i = \phi_s e^{-\phi_d \cdot i} + 1 \quad (7)$$

Context-to-feature associations update as:

$$\Delta M^{CF} = \lambda \phi_i f_i c_i^\top \quad (8)$$

Here, λ scales context-to-feature learning, while ϕ_s and ϕ_d define the size and decay of the primacy boost. Retrieval uses the current context state to assign support to candidate items, sample from the resulting competition, and terminate the sampling sequence. During context-guided retrieval, current context cues encoded items through M^{CF} :

$$a_i = [M^{CF} c]_i \quad (9)$$

The choice sensitivity τ controls how strongly support differences affect choice among candidates:

$$A_i = a_i^\tau \quad (10)$$

The probability of sampling item i , conditional on continuing retrieval, follows a Luce choice rule over currently unsampled items:

$$P(i \mid \text{continue}) = \frac{A_i}{\sum_k A_k} \quad (11)$$

At each retrieval step, the model may terminate with probability $P_{\text{stop}}(j)$ or sample an item with probability $(1 - P_{\text{stop}}(j)) \cdot P(i \mid \text{continue})$. Termination probability grows with retrieval-attempt position:

$$P_{\text{stop}}(j) = \theta_s e^{j\theta_r} \quad (12)$$

Here, j is the number of items already sampled, θ_s sets the baseline termination probability, and θ_r controls its growth. In simulations, this value is bounded to remain a valid probability. After a sampled item r_j , context is updated through the same feature-to-context pathway:

$$c_{j+1} = \rho_{\text{rec},j} c_j + \beta_{\text{rec}} \widehat{M^{FC} f_{r_j}} \quad (13)$$

This sampled-item update drives context toward the sampled item’s encoding neighborhood, biasing subsequent retrievals toward nearby items and producing temporal organization in retrieval. Encoding operations use the context update and learning equations together, whereas reminders, sampled items, and supplied film items use the context update without adding a new recallable event.

Retrieval Control

Retrieval-control settings alter the starting state and selection process used by the same context-to-feature retrieval operation. Unguided retrieval and deliberate recall use the same retrieval-support and choice rules in Equation 9 and Equation 11. They differ in the control settings applied to the retrieval state and sampling process. The first deliberate-recall control setting is start-of-list context reinstatement:

$$c_{\text{start}} = \rho_{\text{start}} c_N + \beta_{\text{start}} c_0 \quad (14)$$

Here, ρ_{start} is calculated from Equation 5 using β_{start} . Simulation settings can use the fitted β_{start} value directly or multiply it to vary the degree of start-of-list context reinstatement. The second control setting is retrieval selectivity, implemented by scaling τ . Higher τ makes selection more sensitive to differences in candidate support. This setting protects film recall only when the retrieval context already gives film items stronger support than task competitors. The third control setting is target monitoring after an off-target item is sampled. Let $t_i = 1$ for film items and $t_i = 0$ otherwise, and define $\kappa_i = t_i + (1 - t_i)\kappa$. Target monitoring modifies the sampled-item context update in Equation 13. With target monitoring, the context update after sampled item i becomes:

$$c_{j+1} = \rho_{\text{rec},j} c_j + \kappa_i \beta_{\text{rec}} \widehat{M^{FC} f_i} \quad (15)$$

When $\kappa < 1$, sampled off-target candidates have less influence on the next context state

than sampled film items. This convention is used for deliberate recall when simulations manipulate rejected-sample updating. Tests that supply a film item are represented as cue-driven context reinstatement using the same feature-to-context input operation as Equation 3:

$$c_{\text{film}}^{\text{IN}} = M^{FC} f_{\text{film}} \quad (16)$$

Supplied film items therefore use the same feature-to-context operation as sampled and reminded items.

Source Context

The source-context pathway adapts the emotional/source-context machinery of eCMR and CMR3 to the present trauma-film setting. In published emotional CMR models, emotional features update a source-context subregion and can form item-context associations (Cohen & Kahana, 2022; Talmi et al., 2019). Here, we use the same architectural role for a simplified film-source dimension: a non-temporal context feature that can organize recall when shared across items (Cohen & Kahana, 2022; Polyn et al., 2009a; Talmi et al., 2019). In the present simulations, this pathway is represented by a simplified source feature vector s_i with a marked film/source feature and a neutral/default feature. Film items are assigned the marked source feature, whereas break, task, and filler items are assigned the neutral/default feature unless a later source-overlap manipulation states otherwise. The source feature vector initializes M_{emot}^{FC} so that an item can cue its source-context components:

$$M_{\text{emot}}^{FC}(\cdot, i) = (1 - \gamma)s_i \quad (17)$$

Source-context-to-feature support begins at zero:

$$M_{\text{emot}}^{CF}(i, \cdot) = 0 \quad (18)$$

Thus, source context can update during experience before it contributes retrieval support. Source-context components use the same context-update rule in Equation 4 with source-context

drift rate β_{emot} . Let η_s denote baseline source-context-to-feature learning and let $\phi_{emot,i}$ denote added learning for item i in the marked source pathway. With the binary vector above, $\phi_{emot,i} = \phi_{emot}[s_i]_1$, so film items assigned the marked source feature receive ϕ_{emot} and neutral/default items receive 0. When source-context learning is engaged, the source components of context receive additional context-to-feature learning:

$$\Delta M_{emot}^{CF} = \lambda \phi_i (\eta_s + \phi_{emot,i}) f_i c_{emot,i}^T \quad (19)$$

Here, $c_{emot,i}$ denotes the source components of context. Source-context support enters retrieval through learned source context-to-feature associations and contributes to the same competition defined in Equation 9. Because η_s is nonzero in the current simulations, neutral/default items can weakly bind the source context active during encoding without being assigned the marked film/source feature. This lets task items inherit some influence from reinstated film-source context when the task is encoded after a film reminder, while still preserving a distinction between marked film items and neutral task competitors.

At retrieval onset, all current simulations include a fixed partial reinstatement of the marked film-source context. This fixed source-context orientation represents the test setting orienting retrieval toward the film episode at the level of source context. It is held constant across unguided and deliberate retrieval and is separate from the start-of-list context reinstatement manipulated in Simulation 2. Task items are assigned the marked source feature only in later source-overlap variants. When film and task items share the marked source feature and source context-to-feature learning is enabled, Equation 19 can amplify competition beyond temporal proximity alone.

These equations define the item-context architecture used in all simulations. During retrieval, the model samples one candidate at a time until stopping. Task competitors reduce film recall when they are sampled instead of film items, using retrieval attempts that could otherwise have returned film material.

Simulations

The simulations instantiate the retrieved-context account in a controlled trauma-film analogue: a laboratory task in which distressing film material is used to study memory processes relevant to responses to real-life trauma (Holmes & Bourne, 2008; James et al., 2016). The simulations are organized into three linked groups: a core context-binding and retrieval-control demonstration, cue and test-format consequences of bidirectional item-context associations, and task-property boundary conditions. Simulation 1 shows how a post-reminder task attenuates film recall when task encoding is strong and occurs in context states that overlap with the film trajectory. Simulation 2 adds the intentionality manipulation by comparing unguided retrieval with deliberate recall and then decomposing the control settings that protect film recall during deliberate retrieval. Simulation 3 asks how recall changes when the test periodically supplies film cues during retrieval, measuring both overall film recall and the film-recall gap between unguided and deliberate retrieval. Simulation 4 compares context-to-item recall with supplied-item access formats. Simulation 5 examines which task properties make the post-film task effective, including task encoding strength, contextual similarity, and source-context overlap.

Methodology

The simulations require default values for context drift, associative learning, retrieval choice, and stopping before the trauma-film manipulations are applied. Ideally, these values would be estimated from a selective interference experiment that implements the full paradigm modeled here: film encoding, reminder or no-reminder conditions, an interference task, a later retrieval test, and an observed dissociation between unguided and deliberate retrieval measures. Such a dataset would also need trial-by-trial recall sequences that identify which film, task, and non-film candidates entered the retrieval stream and in what order, so the model could estimate retrieval competition rather than only final film memory scores. Existing trauma-film studies define the target phenomenon, but they typically do not provide this model-fitting structure. We therefore use an immediate affective free-recall dataset as a provisional anchor for core eCMR dynamics before applying the selective interference manipulations in simulation. The anchor

dataset included participants who reported concern about the war in Ukraine and studied and recalled lists of war-related and neutral photographs ([Dupertuys et al., 2026](#)). The fitted dataset included 123 participants, four lists per participant, 492 immediate recall sequences, and 3,804 scored recall events. The fit provides an externally estimated eCMR regime for deliberate context-guided recall of affective and neutral photographs. The fitted parameters set the starting encoding, retrieval, and stopping dynamics before the selective interference manipulations are applied as phase-specific encoding or retrieval-control settings. In these simulations, film items reinstate the marked source feature into source context, neutral/default events can weakly bind active source context, and retrieval begins with partial film-source reinstatement. To accommodate the difference in total encoded items between the immediate-recall paradigm and the current simulations, the primacy profile is also moderated relative to the immediate-recall fit so that the film sequence is not dominated by an unrealistically steep first-item advantage. [Table 2](#) gives the default numerical values used in the current simulations.

The current simulations update associations after context integration and disallow repeated sampling of the same item. Across simulations, each condition sets phase-specific drift, learning, and retrieval-control parameters, then runs the model through encoding and retrieval. Film, break/buffer, task, and filler items are encoded by the same CMR operations defined above. When modeled as cue events, reminder cues and supplied film items update context through [Equation 3](#) and [Equation 4](#) without adding new recallable items. Simulations that supply item information in a different test format state that separately. Each simulated trial contains 16 film items, 16 break/buffer items, 16 task items, and 16 filler items unless otherwise noted. Equal phase lengths make redistribution across film, task, and filler periods directly comparable. Retrieval allows up to 48 sampling attempts. In context-to-item recall simulations, all encoded film, break/buffer, task, and filler items can enter the retrieval competition. The simulated retrieval sequence records sampled candidates rather than only final accepted responses. This convention lets target monitoring count an off-target sample as a retrieval attempt, attenuate its context update, and still allow search to continue.

A parameter sweep involves rerunning the full encode-then-retrieve simulation across planned values of a manipulated parameter. For each simulation condition and manipulated parameter value, the model first completes the phase sequence, then enters retrieval and samples recallable items using Equation 11 and Equation 12. This procedure is repeated across simulated trials within each condition/value cell. When figures plot recall across the encoded stream, they summarize retrieval after the full phase sequence has been encoded. Recall curves are estimated across simulated trials by scoring which encoded items appeared in each retrieval sequence. Serial-position curves plot the proportion of trials in which each encoding position was retrieved, and phase summaries average those proportions within phase. Thus, position and phase labels identify the encoding positions, whereas recall probabilities refer to subsequent retrieval.

Simulation 1: Task Encoding and Contextual Overlap

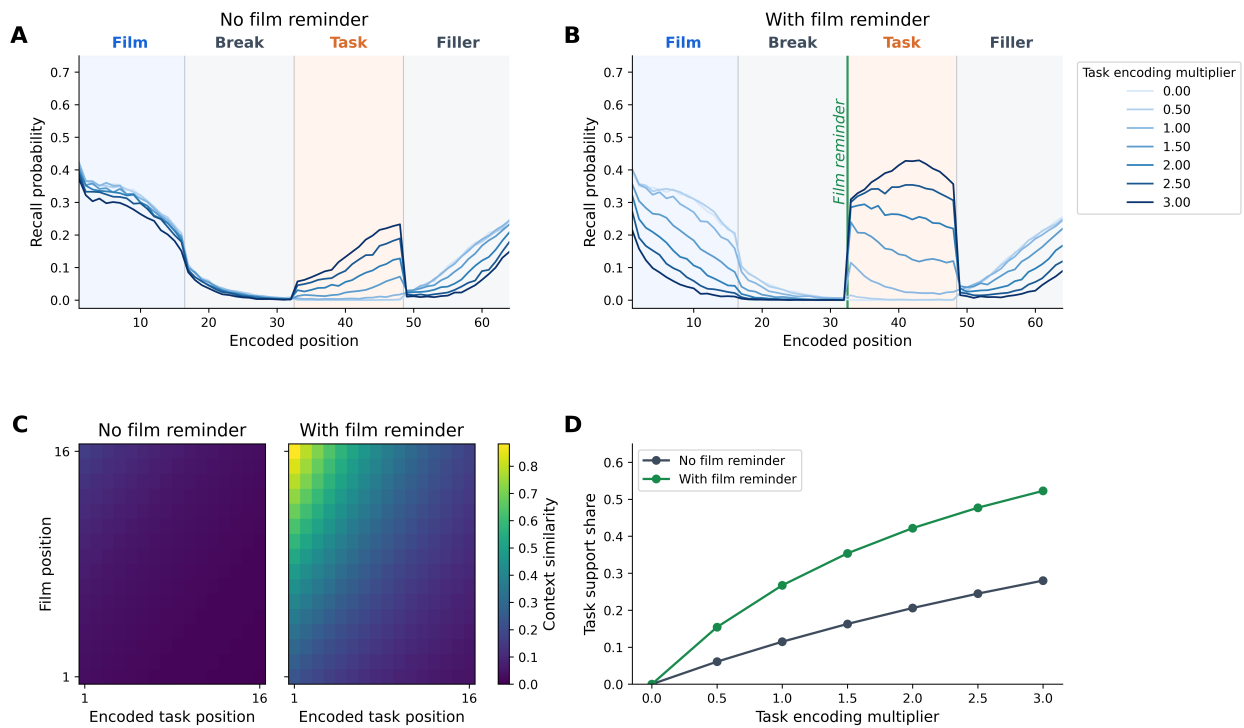
Simulation 1 demonstrates how a post-film task can reduce later film recall when task events are encoded in context states that overlap with the film sequence. To separate task encoding strength from reminder-driven context reinstatement, the simulation runs both the no-reminder and with-reminder conditions across the same set of task encoding strengths. We express task encoding strength as a task encoding multiplier that scales task-phase context-to-feature learning in Equation 8: 0 removes new task context-to-feature learning, 1 applies the baseline fitted rate, and values above 1 strengthen task item-context binding. In the no-reminder condition, the reminder-start and film-cue drift scales are set to zero. In the with-reminder condition, context first shifts toward the start state and then receives sequential film reminder cues before task encoding. Reminder cues update context without applying Equation 6 or Equation 8 and without adding new recallable items. Retrieval settings are held fixed, so differences in recall reflect the item-context associations formed before retrieval rather than changes in retrieval control. Because each film position is encoded with a different temporal-context state, the model predicts a serial-position pattern, not only an overall change in film recall. Task items should compete most with the film positions whose encoding contexts are most similar to the contexts active during task encoding. Table 3 summarizes the phase structure and settings used for Simulation 1. For each

reminder condition and task encoding multiplier, serial-position curves estimate the probability that each encoded position appears at least once in the later sampled retrieval sequence. As described above, these serial-position curves sort retrieval outcomes by encoded position. Figure 2 also inspects two internal model quantities that help explain those recall curves. The context-similarity analysis asks how similar each film-encoding context is to the contexts active while task items are encoded, using cosine similarity over unit-normalized temporal context states. The support-allocation analysis asks how much retrieval support would go to task items when the model is probed with an average film context after task encoding. It forms that probe from the centroid of film encoding context states, applies it to M^{CF} , and computes task support divided by total film-plus-task support.

Figure 2

Reminder-driven context overlap makes task encoding strength consequential for film recall.

Simulation 1 demonstrates how task items can become competitors for film recall by varying the task encoding multiplier with or without a sequential film reminder before task encoding. (A-B) Recall is relatively stable without a reminder but shifts from film to task positions when a reminder precedes task encoding. (C) The reminder makes task-encoding states more similar to a position-resolved portion of the film context trajectory. (D) The same reminder increases task support from a film-context probe as task encoding strengthens.



Without a film reminder, increasing the task encoding multiplier produces only modest loss of film recall and only modest gain in task recall, because task items become associated with later context states reached after drift away from the film. With a film reminder, the same task-encoding manipulation produces a larger drop in film-position recall and a larger rise in task-position recall. The with-reminder condition therefore shows that task encoding strength matters most when task items are encoded in a film-related context state. At the lowest and highest multiplier values, mean film recall in the no-reminder condition changes from 0.318 at

task encoding multiplier 0.0 to 0.268 at multiplier 3.0, while mean task recall changes from 0.000 to 0.143. In the with-reminder condition, mean film recall changes from 0.306 to 0.064, while mean task recall changes from 0.000 to 0.384.

Panels C and D explain why the no-reminder and with-reminder curves diverge. Panel C quantifies temporal-context similarity between the states associated with film items and the states active during task encoding. When a sequential film reminder reinstates film-associated context before task encoding, the task-encoding contexts become more similar to some film-position contexts than to others. Panel D quantifies how much support task items receive from the average film-context probe after task encoding, including the source-context support available at that point in the paradigm. As the task encoding multiplier increases, task items receive a larger share of item support when this temporal/source probe is applied, especially after the film reminder. Across multiplier values, film encoding settings remain unchanged. The loss of film recall comes from stronger support for task items that can be cued by film-associated context. In paradigm terms, the reminder leaves the model partly oriented toward the film as task encoding begins, so task events become linked to film-like temporal context as well as the active film-source state. Task items remain distinct from film items but become increasingly likely to win retrieval from film-related cues. Simulation 2 next examines whether deliberate retrieval control reduces the impact of these task-item associations by reinstating an early-film context state that favors film items over task competitors.

Simulation 2: Control Mechanisms for Deliberate Recall

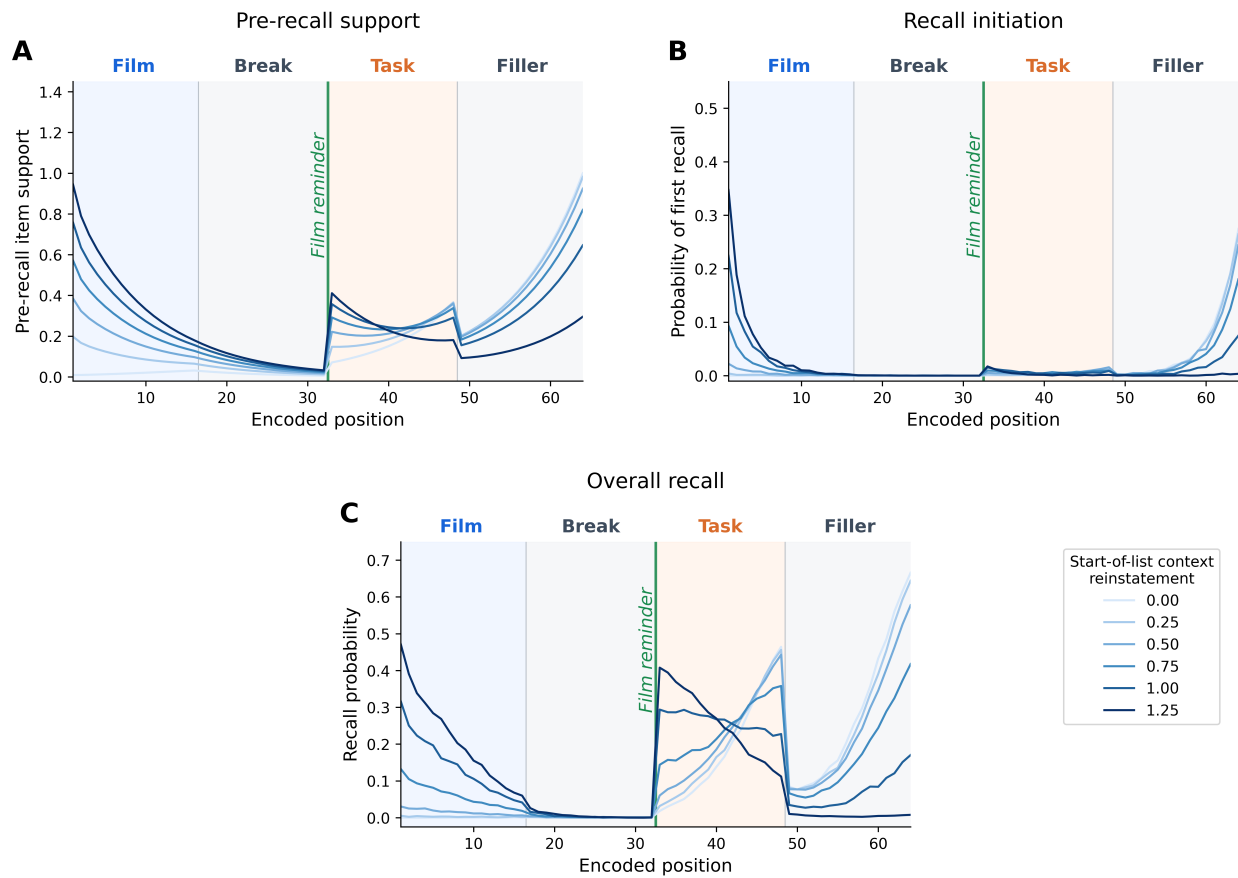
Simulation 2 turns from encoding conditions to retrieval control. The broader Simulation 2 package decomposes deliberate recall into three control settings: start-of-list context reinstatement, retrieval selectivity, and target monitoring. Figure 3 isolates the first setting. In the model, deliberate recall can begin by drifting current context toward the context state from the start of the encoded sequence before the first retrieval attempt. Because the film is encoded first in this paradigm, that start state supports early film items while providing relatively little support for later task or filler items. This manipulation changes the starting point of memory search while

holding the learned film and task associations fixed.

All Simulation 2 figures presented here use the with-reminder task-encoding condition from Simulation 1, with the task encoding multiplier fixed at 2.0. All conditions also include the fixed partial source-context orientation toward the marked film source described above. Thus, “no start-of-list reinstatement” means no deliberate reinstatement of the temporal start state, not the absence of any film-relevant test context. Figure 3 varies the degree of start-of-list context reinstatement. Reinstatement strength controls how strongly retrieval context is pulled toward the start state before the first retrieval attempt. Values are expressed relative to the fitted start drift rate, so 1.0 corresponds to β_{start} in Equation 14.

Figure 3

Start-of-list context reinstatement protects deliberate film recall by shifting the starting retrieval context. Figure 3 isolates this start-state manipulation in the with-reminder task-encoding condition: before the first retrieval attempt, context is drifted toward the start of the encoded sequence. (A) Stronger reinstatement shifts pre-recall support from late filler positions toward early film positions. (B) Recall initiation follows this support shift. (C) The initial shift carries into overall recall, with the largest gains near the beginning of the film sequence.



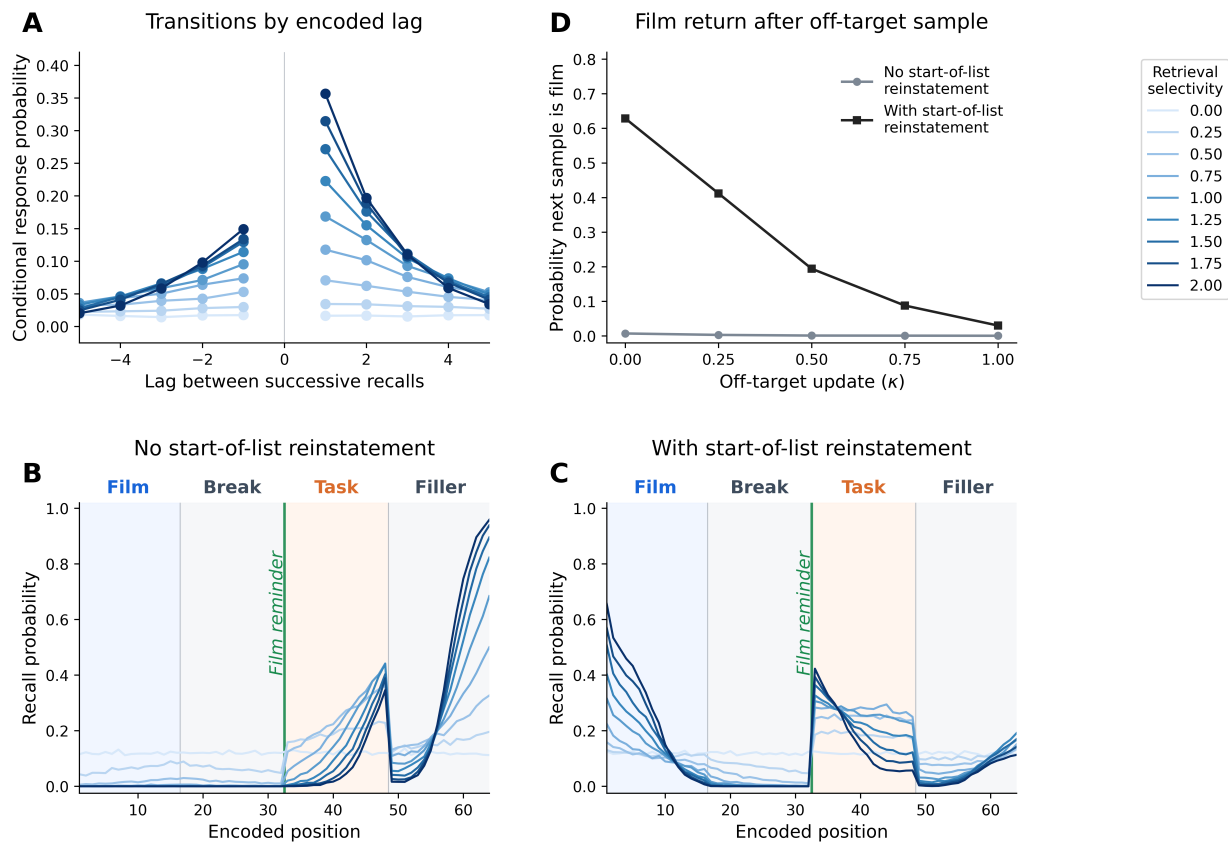
Panel A is a model-state readout: it shows how much each encoded position is supported by the retrieval context immediately before the first retrieval attempt, before the model applies the choice, stopping, or context-updating rules. With no start-of-list context reinstatement, support is concentrated in late filler positions because the retrieval context reflects the post-filler state. As start-of-list context reinstatement increases, support shifts toward early film positions. Panels B

and C summarize sampled retrieval sequences under the same retrieval settings. Panel B shows that this support shift changes recall initiation, with first recalls moving from filler positions when start-of-list context reinstatement is omitted to early film positions as reinstatement increases. Panel C shows that the initiation shift extends beyond the first response. Once early film items are sampled, they reinstate nearby early-film context; gradual context drift then keeps later samples closer to the film portion of the encoded sequence. Under the strong task setting used here, task items remain available to be sampled later in the sequence, so the next step is to examine how selection operates after retrieval has been oriented toward early film context.

The second control setting is retrieval selectivity. This manipulation scales the choice-sensitivity parameter τ in Equation 10, so candidates with stronger current support are selected more disproportionately often. Figure 4 crosses retrieval selectivity with start-of-list context reinstatement while holding fixed the same with-reminder task-encoding condition used in Figure 3. The final panel examines target monitoring through rejected-sample updating. After an off-target candidate is sampled, the model varies how much that candidate updates context before search continues. Sampled off-target candidates still occupy retrieval attempts, but their context update is multiplied by κ , with $\kappa = 1$ giving the normal sampled-item update and $\kappa = 0$ preventing off-target context updating.

Figure 4

Retrieval selectivity amplifies the context already in place, while target monitoring limits drift after off-target samples. Simulation 2 varies retrieval selectivity in the with-reminder task-encoding condition, with or without start-of-list context reinstatement, and then examines rejected-sample updating to show why selective sampling and monitoring depend on the retrieval context already in place. (A) Higher selectivity increases local transitions when retrieval starts from early-film context. (B-C) The same increase favors late non-film positions without start-of-list reinstatement and film positions with it. (D) Reducing off-target context updating helps retrieval return to film only when start-of-list reinstatement has already made film items accessible.



Values near zero in the lag-CRP indicate local transitions; because the Luce choice rule exaggerates differences in candidate support, higher retrieval-selectivity values make recalls cluster more strongly around nearby encoded positions. Panels B and C show that the recall

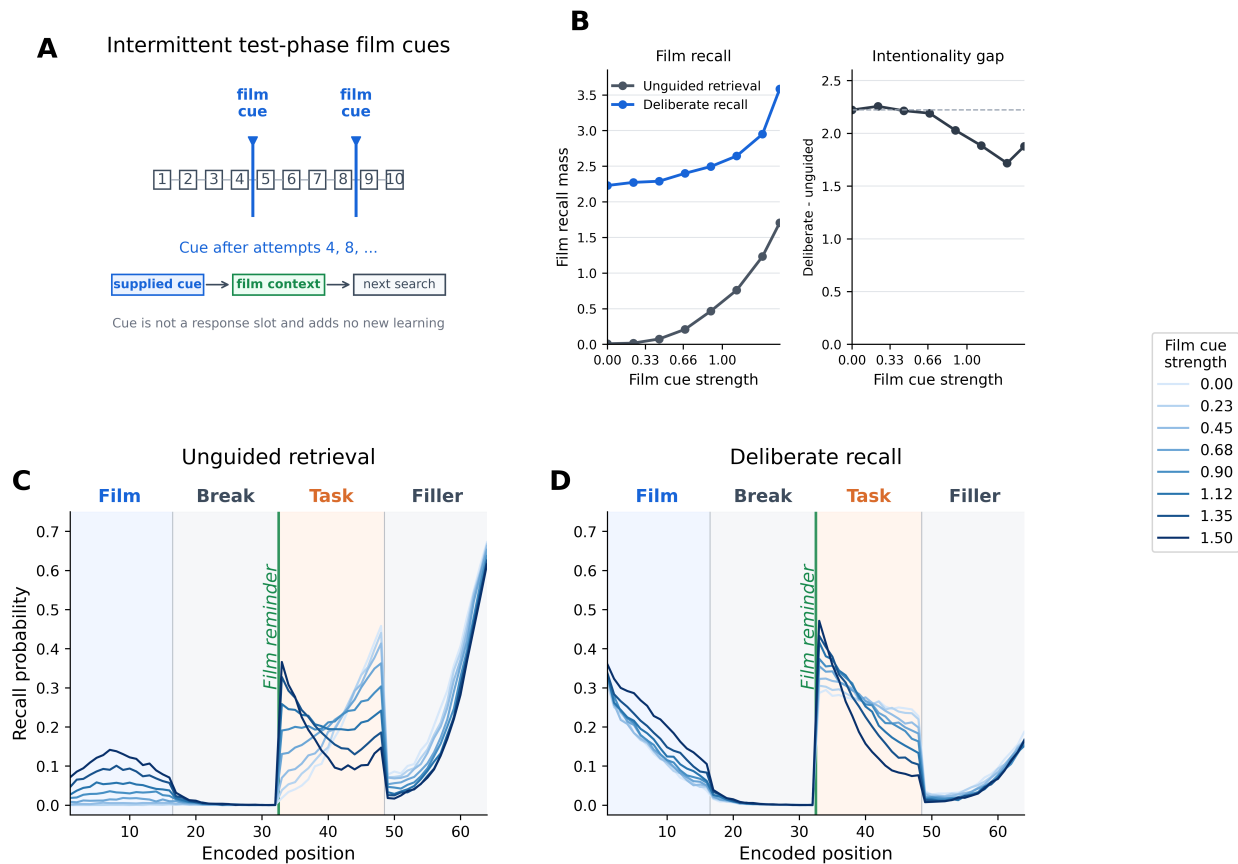
consequences of this local search depend on the retrieval context. When recall begins from the ongoing post-filler context, increasing retrieval selectivity favors the late non-film items most strongly supported by that context. When recall begins after start-of-list context reinstatement, increasing retrieval selectivity favors the early film items supported by the reinstated context. Panel D conditions on transitions where the current sample is off target, retrieval continues, and at least one film item remains unsampled. Without start-of-list context reinstatement, the probability that the next sample is a film item remains near zero across the κ sweep. With start-of-list context reinstatement, lowering κ from 1.0 to 0.0 raises this film-return probability from 0.059 to 0.643. Across Figures 3 and 4, deliberate recall is protected by a sequence of control settings: start-of-list context reinstatement establishes a diagnostic early-film retrieval context, retrieval selectivity amplifies differences in whatever the current context supports, and target monitoring limits the context-updating consequences of sampled off-target candidates. This sequence sets up a contrast with test-phase film cues. Unlike the pre-task film reminder that makes task encoding film-related, Simulation 3 presents film items during retrieval as external cues. Their protective value depends on whether the reinstated context favors film items over task competitors.

Simulation 3: Test-Phase Film Cues and Retrieval Intentionality

Simulation 3 asks whether test-phase film cues have the same protective value as deliberate start-of-list reinstatement. In the model, these cues are implemented as randomly selected film items supplied during retrieval: after every fourth retrieval attempt, the supplied film item updates temporal and source context through its feature-to-context associations, after which ordinary retrieval resumes. The supplied item does not occupy a response slot, does not become unavailable for later recall, and adds no new learning. Cue strength controls how strongly each supplied film item pulls context toward the film state associated with that item. The plotted values express that pull relative to the fitted recall context-update rate β_{rec} . The vertical reference line in Figure 5 marks the point at which a supplied cue would move context by about the same amount as one ordinary encoding update, approximately β_{enc} .

Figure 5

Test-phase film cues boost film recall, but deliberate recall remains advantaged. Simulation 3 compares unguided retrieval with deliberate recall while varying the strength of film cues supplied during retrieval. **(A)** Film cues are supplied after scheduled retrieval attempts, update context, and add no new learning. **(B)** Film recall increases with cue strength, but deliberate recall remains higher than unguided retrieval across weak-to-moderate cue strength. Only stronger cues begin to narrow the gap. **(C-D)** Across the encoded sequence, film cues raise film recall without eliminating task and filler competitors.



Panel B gives the primary test of the Simulation 3 prediction. Weak-to-moderate test-phase film cues increase film recall while leaving the intentionality gap largely intact. For example, increasing film cue strength from 0.00 to 0.675 raises film recall mass from 0.008 to 0.210 under unguided retrieval and from 2.229 to 2.400 under deliberate recall, while the

deliberate-minus-unguided gap changes only from 2.221 to 2.190. This pattern shows that a test-phase film cue can increase film recall without producing the same diagnostic early-film starting context as Simulation 2's start-of-list reinstatement. At stronger cue values, the intentionality gap begins to narrow because cue-driven context reinstatement itself becomes strong enough to guide unguided retrieval toward film items.

Panels C and D show how this effect unfolds over the full encoded sequence. The cue manipulation does not remove the task competitors created by reminder-driven task encoding. Instead, it changes the context available to subsequent search. The result is a graded cue effect: supplied film items can increase film recall, but they reduce selective interference only to the extent that the reinstated context discriminates film targets from task and filler competitors.

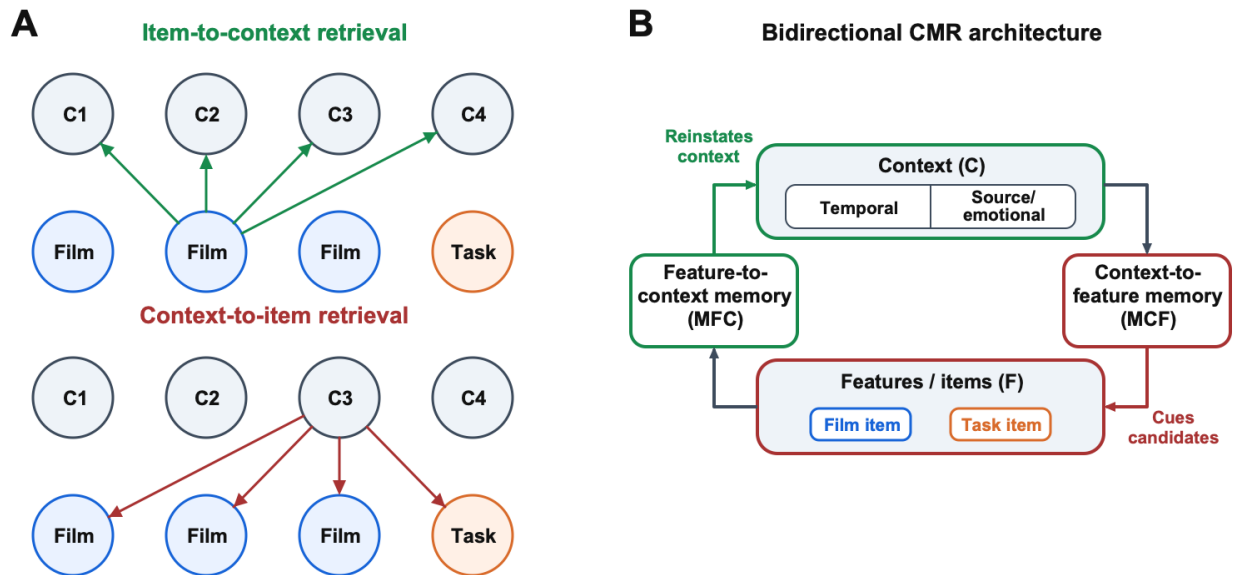
Simulation 4: Context-to-Item and Item-to-Context Test Formats

Simulation 4 extends the cue logic to test format. The relevant architectural distinction is between context-to-item recall, in which film items must win selection from a mixed competitor set, and supplied-item access, in which the test provides item information that can reinstate associated context directly. This comparison does not require treating recognition as fully modeled by CMR. Instead, it asks whether a supplied film item changes vulnerability to context-binding interference because the retrieval problem no longer begins with selecting a film item from context alone.

The predicted contrast follows from the bidirectional item-context loop. When the test begins from context, film and task items compete according to their context-to-feature support. When the test supplies an item, item-to-context associations are engaged first, so performance can be less sensitive to whether a film item would have won an unguided context-to-item competition. This provides a test-format consequence of the same mechanism examined in Simulation 3, without assuming a separate recognition decision model.

Figure 6

Retrieval format changes where interference can enter. *Simulation 4 distinguishes context-guided retrieval from supplied-item retrieval. (A) When retrieval begins from context, the same context state can cue multiple candidates; task items therefore compete with film items when both have been encoded in overlapping contexts. When retrieval begins from a supplied film item, the item reinstates its associated context instead, so the task item is not an equivalent competitor at the point of access. (B) In CMR, this asymmetry corresponds to two retrieval directions: context-to-feature associations cue candidate items, whereas feature-to-context associations reinstate context from a supplied item. Simulation 4 uses this distinction to ask why item-cued or recognition-like tests may show preserved access even when context-guided recall is sensitive to interference.*



Simulation 5: Task Specificity and Source-Context Overlap

Simulation 5 turns to task specificity. In the model, “visuospatial” and “comparison” are not primitive task labels. A post-film task becomes effective when its items are strong competitors, when they are encoded in temporal-context states that overlap with the film trajectory, or when they share a source-context state with film items. These factors can vary independently, so a task label is interpretable only through its consequences for competitor strength and contextual overlap.

The task-specificity simulations therefore decompose three candidate routes to interference. Increasing the task encoding multiplier changes how strongly task items win once cued. Changing temporal overlap alters whether task items are cued alongside film items. Changing source-context overlap tests whether non-temporal source matching can amplify competition beyond temporal proximity alone. The predicted outcomes are film access, task-competitor retrieval, and retrieval outcomes by encoded phase or position.

General Discussion

References

- Asselbergs, J., van Bentum, J., Riper, H., Cuijpers, P., Holmes, E. A., & Sijbrandij, M. (2023). A systematic review and meta-analysis of the effect of cognitive interventions to prevent intrusive memories using the trauma film paradigm. *Journal of Psychiatric Research*, 159, 116–129. <https://doi.org/10.1016/j.jpsychires.2023.01.028>
- Baddeley, A. D., & Andrade, J. (2000). Working memory and the vividness of imagery. *Journal of Experimental Psychology: General*, 129(1), 126–145. <https://doi.org/10.1037/0096-3445.129.1.126>
- Bonsall, M. B., & Holmes, E. A. (2023). Temporal dynamics of trauma memory persistence. *Journal of the Royal Society Interface*, 20(203), 20230108. <https://doi.org/10.1098/rsif.2023.0108>
- Boyle, A., & Brown, S. A. B. (2025). Why might animals remember? A functional framework for episodic memory research in comparative psychology. *Learning & Behavior*, 53, 14–30. <https://doi.org/10.3758/s13420-024-00645-0>
- Brewin, C. R. (2014). Episodic memory, perceptual memory, and their interaction: Foundations for a theory of posttraumatic stress disorder. *Psychological Bulletin*, 140(1), 69–97. <https://doi.org/10.1037/a0033722>
- Brewin, C. R., Dalgleish, T., & Joseph, S. (1996). A dual representation theory of posttraumatic stress disorder. *Psychological Review*, 103(4), 670–686. <https://doi.org/10.1037/0033-295X.103.4.670>

- Brewin, C. R., Gregory, J. D., Lipton, M., & Burgess, N. (2010). Intrusive images in psychological disorders: Characteristics, neural mechanisms, and treatment implications. *Psychological Review*, 117(1), 210–232. <https://doi.org/10.1037/a0018113>
- Cohen, R. T., & Kahana, M. J. (2022). A memory-based theory of emotional disorders. *Psychological Review*, 129(4), 742–776. <https://doi.org/10.1037/rev0000334>
- Deepröse, C., Zhang, S., DeJong, H., Dalgleish, T., & Holmes, E. A. (2012). Imagery in the aftermath of viewing a traumatic film: Using cognitive tasks to modulate the development of involuntary memory. *Journal of Behavior Therapy and Experimental Psychiatry*, 43(2), 758–764. <https://doi.org/10.1016/j.jbtep.2011.10.008>
- Dupertuys, J., Shulhan, A., & Talmi, D. (2026). *Ukraine Memory experiment data set*.
- Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. *Behaviour Research and Therapy*, 38(4), 319–345. [https://doi.org/10.1016/S0005-7967\(99\)00123-0](https://doi.org/10.1016/S0005-7967(99)00123-0)
- Holmes, E. A., & Bourne, C. (2008). Inducing and modulating intrusive emotional memories: A review of the trauma film paradigm. *Acta Psychologica*, 127(3), 553–566. <https://doi.org/10.1016/j.actpsy.2007.11.002>
- Holmes, E. A., Brewin, C. R., & Hennessey, R. G. (2004). Trauma films, information processing, and intrusive memory development. *Journal of Experimental Psychology: General*, 133(1), 3–22. <https://doi.org/10.1037/0096-3445.133.1.3>
- Holmes, E. A., James, E. L., Coode-Bate, T., & Deepröse, C. (2009). Can playing the computer game Tetris reduce the build-up of flashbacks for trauma? A proposal from cognitive science. *PLoS ONE*, 4(1), e4153. <https://doi.org/10.1371/journal.pone.0004153>
- Holmes, E. A., James, E. L., Kilford, E. J., & Deepröse, C. (2010). Key steps in developing a cognitive vaccine against traumatic flashbacks: Visuospatial Tetris versus verbal Pub Quiz. *PLoS ONE*, 5(11), e13706. <https://doi.org/10.1371/journal.pone.0013706>
- Horsch, A., Vial, Y., Favrod, C., Harari, M. M., Blackwell, S. E., Watson, P., Iyadurai, L., Bonsall, M. B., & Holmes, E. A. (2017). Reducing intrusive traumatic memories after emergency caesarean section: A proof-of-principle randomized controlled study. *Behaviour Research*

and Therapy, 94, 36–47. <https://doi.org/10.1016/j.brat.2017.03.018>

Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context.

Journal of Mathematical Psychology, 46(3), 269–299.

<https://doi.org/10.1006/jmps.2001.1388>

Howard, M. W., Youker, T. E., & Venkatadass, V. S. (2008). The persistence of memory:

Contiguity effects across hundreds of seconds. *Psychonomic Bulletin & Review*, 15(1), 58–63.

Hupbach, A., Hardt, O., Gomez, R., & Nadel, L. (2008). The dynamics of memory:

Context-dependent updating. *Learning & Memory*, 15(8), 574–579.

<https://doi.org/10.1101/lm.1022308>

Iyadurai, L., Blackwell, S. E., Meiser-Stedman, R., Watson, P. C., Bonsall, M. B., Geddes, J. R.,

Nobre, A. C., & Holmes, E. A. (2018). Preventing intrusive memories after trauma via a brief intervention involving Tetris computer game play in the emergency department: A proof-of-concept randomized controlled trial. *Molecular Psychiatry*, 23(3), 674–682.

<https://doi.org/10.1038/mp.2017.23>

Iyadurai, L., Visser, R. M., Lau-Zhu, A., Porcheret, K., Horsch, A., Holmes, E. A., & James, E. L.

(2019). Intrusive memories of trauma: A target for research bridging cognitive science and its clinical application. *Clinical Psychology Review*, 69, 67–82.

<https://doi.org/10.1016/j.cpr.2018.08.005>

James, E. L., Bonsall, M. B., Hoppitt, L., Tunbridge, E. M., Geddes, J. R., Milton, A. L., &

Holmes, E. A. (2015). Computer game play reduces intrusive memories of experimental trauma via reconsolidation-update mechanisms. *Psychological Science*, 26(8), 1201–1215.

<https://doi.org/10.1177/0956797615583071>

James, E. L., Lau-Zhu, A., Clark, I. A., Visser, R. M., Hageraars, M. A., & Holmes, E. A. (2016).

The trauma film paradigm as an experimental psychopathology model of psychological trauma: Intrusive memories and beyond. *Clinical Psychology Review*, 47, 106–142.

<https://doi.org/10.1016/j.cpr.2016.04.010>

Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24(1),

103–109.

Kanstrup, M., Kontio, E., Geranmayeh, A., Olofsdotter Lauri, K., Moulds, M. L., & Holmes, E.

A. (2021). A single case series using visuospatial task interference to reduce the number of visual intrusive memories of trauma with refugees. *Clinical Psychology & Psychotherapy*, 28(1), 109–123. <https://doi.org/10.1002/cpp.2489>

Kessler, H., Schmidt, A.-C., James, E. L., Blackwell, S. E., von Rauchhaupt, M., Harren, K.,

Kehyayan, A., Clark, I. A., Sauvage, M., Herpertz, S., Axmacher, N., & Holmes, E. A. (2020). Visuospatial computer game play after memory reminder delivered three days after a traumatic film reduces the number of intrusive memories of the experimental trauma. *Journal of Behavior Therapy and Experimental Psychiatry*, 67, 101454.

<https://doi.org/10.1016/j.jbtep.2019.01.006>

Kindt, M., Soeter, M., & Vervliet, B. (2009). Beyond extinction: Erasing human fear responses and preventing the return of fear. *Nature Neuroscience*, 12(3), 256–258.

<https://doi.org/10.1038/nm.2271>

Kragel, J. E., Morton, N. W., & Polyn, S. M. (2015). Neural activity in the medial temporal lobe reveals the fidelity of mental time travel. *Journal of Neuroscience*, 35(7), 2914–2926.

<https://doi.org/10.1523/JNEUROSCI.3378-14.2015>

Krans, J., Näring, G., Becker, E. S., & Holmes, E. A. (2009). Intrusive trauma memory: A review and functional analysis. *Applied Cognitive Psychology*, 23(8), 1076–1088.

<https://doi.org/10.1002/acp.1611>

Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2019). Intrusive memories and voluntary memory of a trauma film: Differential effects of a cognitive interference task after encoding. *Journal of Experimental Psychology: General*, 148(12), 2154–2180. <https://doi.org/10.1037/xge0000598>

Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2021). Selectively interfering with intrusive but not voluntary memories of a trauma film: Accounting for the role of associative memory.

Clinical Psychological Science, 9(6), 1128–1143. <https://doi.org/10.1177/2167702621998315>

Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2015). Expanding the scope of memory search:

- Modeling intralist and interlist effects in free recall. *Psychological Review*, 122(2), 337–363.
<https://doi.org/10.1037/a0039036>
- Morton, N. W., & Polyn, S. M. (2016). A predictive framework for evaluating models of semantic organization in free recall. *Journal of Memory and Language*, 86, 119–140.
<https://doi.org/10.1016/j.jml.2015.10.002>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009a). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review*, 116(1), 129–156.
<https://doi.org/10.1037/a0014420>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009b). Task context and organization in free recall. *Neuropsychologia*, 47(11), 2158–2163.
<https://doi.org/10.1016/j.neuropsychologia.2009.02.013>
- Sederberg, P. B., Gershman, S. J., Polyn, S. M., & Norman, K. A. (2011). Human memory reconsolidation can be explained using the temporal context model. *Psychonomic Bulletin & Review*, 18(3), 455–468. <https://doi.org/10.3758/s13423-011-0086-9>
- Sederberg, P. B., Howard, M. W., & Kahana, M. J. (2008). A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4), 893–912.
<https://doi.org/10.1037/a0013396>
- Talmi, D., Lohnas, L. J., & Daw, N. D. (2019). A retrieved context model of the emotional modulation of memory. *Psychological Review*, 126(4), 455–485.
<https://doi.org/10.1037/rev0000132>
- Varma, M. M., Zeng, S., Singh, L., Holmes, E. A., Huang, J., Chiu, M. H., & Hu, X. (2024). A systematic review and meta-analysis of experimental methods for modulating intrusive memories following lab-analogue trauma exposure in non-clinical populations. *Nature Human Behaviour*, 8(10), 1968–1987. <https://doi.org/10.1038/s41562-024-01956-y>
- Wessel, I., Krans, J., Albers, C. J., Chauhan, N., Ferreira de Sá, D. S., Hauck, A., Jaswetz, L., Michael, T., Nixon, R. D. V., Pearson, D. G., Slattery, I., Smeets, T., Takarangi, M. K. T., Willems, T. R. A., & van Schie, K. (2025). Evidence that Tetris reduces immediate but not

subsequent daily intrusions of a trauma film: A multilab replication study. *Collabra:*

Psychology, 11(1), 130791. <https://doi.org/10.1525/collabra.130791>

Zaromb, F. M., Howard, M. W., Dolan, E. D., Sirotin, Y. B., Tully, M., Wingfield, A., & Kahana,

M. J. (2006). Temporal associations and prior-list intrusions in free recall. *Journal of*

Experimental Psychology: Learning, Memory, and Cognition, 32(4), 792–804.

<https://doi.org/10.1037/0278-7393.32.4.792>

Zhou, C. Y., Talmi, D., Daw, N. D., & Mattar, M. G. (2025). Episodic retrieval for model-based evaluation in sequential decision tasks. *Psychological Review*, 132(1), 18–49.

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Table 1

Model parameters and roles in the model specification. Base parameters are the unscaled values used by the model equations before simulation-specific multipliers are applied. A phase is a segment of the simulated paradigm, such as film encoding, intervening delay, reminder, task encoding, post-task delay, or retrieval.

Symbol	Name	Role
β_{enc}	encoding drift rate	Context drift during study
β_{start}	start drift rate	Drift toward start-of-list context at recall onset
β_{rec}	recall drift rate	Context drift after recall
γ	feature-to-context learning rate	Learning rate for M^{FC}
λ	context-to-feature learning scale	Baseline or phase-scaled learning rate for M^{CF}
α	shared support	Pre-experimental shared support in M^{CF}
δ	self support	Pre-experimental self-support in M^{CF}
ϕ_s	primacy scale	Initial boost to context-to-feature learning
ϕ_d	primacy decay	Decay rate of primacy boost
τ	choice sensitivity	Sensitivity of choice to support differences
β_{emot}	source-context drift rate	Source context drift
ϕ_{emot}	source learning scale	Learning scale for the marked source feature
κ	target-monitoring scale	Reduced context updating after sampled off-target candidates when target monitoring is enabled
θ_s	stop scale	Baseline stopping probability

Table 2

Default numerical parameter values. Values are from a pooled eCMR fit to the immediate free-recall sequences from Dupertuys et al. (2026), with α fixed to zero, except for the source-context and primacy settings described in the text.

Symbol	Value	Role
β_{enc}	0.440	Encoding drift rate
β_{start}	0.766	Start drift rate
β_{rec}	0.678	Recall drift rate
γ	0.468	Feature-to-context learning rate
δ	0.00022	Pre-experimental self-support
α	0	Pre-experimental shared support
ϕ_s	0.100	Primacy scale used in simulations
ϕ_d	1.500	Primacy decay used in simulations
τ	3.562	Choice sensitivity
β_{emot}	0.696	Source-context drift rate
ϕ_{emot}	0.130	Learning scale for the marked source feature
η_s	0.050	Baseline source-context-to-feature learning
$\beta_{film-source}$	0.500	Fixed film-source reinstatement at retrieval onset
θ_s	0.0123	Stop scale
θ_r	0.372	Stop growth

Table 3

Phase structure and settings for Simulation 1. Scale values multiply the corresponding base rates or learning strengths.

Phase or setting	Current value	Role
Film items	16	Target episode
Break/buffer items	16	Intervening non-film experience before reminder/task
Film reminder cues	16	Sequential film cues, no new learning
Task items	16	Potential competitors
Filler items	16	Post-task non-film experience
Maximum sampling attempts	48	Upper bound on sampled retrieval attempts
Simulated trials per condition	5000	For each reminder condition and task encoding multiplier
Break and filler drift/learning scales	1.0	Baseline non-film encoding settings
No-reminder drift scales	0.0	Omits reminder-start and reminder-cue reinstatement
With-reminder drift scales	1.0	Applies reminder-start and film cue reinstatement
Task encoding multiplier	0.0–3.0 by 0.5	Multiplies task-phase context-to-feature learning during task encoding
Film source marking	On	Film items carry the marked source feature
Task source marking	Off	Task items use the